Yield-Constrained Agrivoltaic Potential of India Under Current and Mid-Century Climate

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Key Points:

• If crops must keep 80% of their yield, Indian farmland could host 22.6 to 39.2 terawatts peak of elevated solar

• Allowed panel density is about twice as high for cereals as for legumes, and low fertiliser use makes crops look more shade tolerant

• Mid-century climate barely changes this capacity, but raised frames need a tariff top-up that crop income cannot cover

Abstract

Growing crops under solar panels could ease the competition between food and energy for land, but national estimates of agrivoltaic potential assume fixed power densities and do not test whether crops keep their yield. To our knowledge, this is the first yield-constrained national potential for India. A strip-resolved light model for three array layouts drives crop models of wheat, chickpea, pearl millet and groundnut on 2977 grid cells, under five downscaled climate models for 1995–2014 and 2041–2060 and with nitrogen calibrated to observed state yields. For each cell we found the largest ground coverage ratio (GCR) that keeps relative yield at or above 0.66, 0.80 or 0.90, the thresholds of German, Japanese and French rules. With the lower of calibrated and recommended nitrogen, cropland with slopes up to 5% could host 39.2 terawatts peak (TWp) of east-west elevated arrays at 80% relative yield on 39.7 million hectares (60.9 petawatt-hours per year). The crop models were more shade-tolerant than field observations; rescaled to them, capacity is 22.6 TWp, so we report 22.6–39.2 TWp as a technical potential, not a forecast. Feasible GCR averaged 0.52 for cereals but 0.29 for legumes, and low nitrogen hid part of the shade loss. Under the high-emission pathway SSP5-8.5, capacity changed by −1% and energy by −5%, through lower irradiance. Screening costs show that elevated frames need a tariff adder of Indian

rupees (Rs) 0.49–0.85 per kWh, far above the crop value protected, so support should be tied to a yield threshold.

Plain Language Summary

Solar farms compete with food crops for land. Panels can be raised above a field so that both share the sunlight, but a crop only does well if enough light gets through. We built a computer model of the light under three panel layouts and linked it to crop models for wheat, chickpea, pearl millet and groundnut across India. For each place we asked how many panels fit if the crop must still give at least 80% of its normal harvest. Under that rule, Indian cropland could host between 22.6 and 39.2 terawatts of solar capacity, a technical figure that ignores grid, finance and land ownership. Wheat and pearl millet tolerate about twice as many panels as chickpea and groundnut, and crops grown with little fertiliser seem to cope with shade better than they really would. Climate change by mid-century barely changes how many panels fit, but it slightly reduces the electricity they produce. Raised frames cost more than ground-mounted panels, so farmers would need a higher electricity price, a gap that the value of the crop cannot cover. Linking any support to a yield rule would keep the land in food production.

1 Introduction

India has committed to drawing half of its installed electricity capacity from non-fossil sources by 2030 (Government of India, 2022), and solar power carries most of that growth, with more than 160 gigawatts (GW) installed by mid-2026 (Ministry of New and Renewable Energy, 2026b). Utility-scale photovoltaics (PV) need flat, sunny and accessible land, which in India is mostly farmland. Agriculture still employs about 46% of the country's workers (National Statistics Office, 2024), so converting cropland to PV trades food and rural income for power, and croplands are among the most productive places for PV worldwide (Adeh et al., 2019). Worldwide, an estimated 13–16% of ground-mounted PV is already associated with a loss of cropland (Curioni et al., 2025). Growing crops under the panels, known as agrivoltaics (APV), is the route that avoids choosing between food and energy.

Early field and modelling work showed that the combined output of food and electricity can exceed that of separate fields (Dupraz et al., 2011; Valle et al., 2017), and in drylands shade can lower plant water stress (Barron-Gafford et al., 2019). The crop

response, however, is not uniform. A meta-analysis of shading experiments found that some crops lose little yield under moderate shade while others, including grain legumes, lose yield almost in proportion to the light removed (Laub et al., 2022). Evidence from India is still sparse: it covers grape farms (Malu et al., 2017), the experimental system at Jodhpur in the hot arid zone (Santra et al., 2021; Poonia et al., 2022) and wheat under arrays of different density (Prakash et al., 2023). A layout that suits one crop can therefore fail another, and APV potential cannot be judged without the crop below it.

For this reason, rules that define APV set a minimum crop yield. Germany requires at least 66% of the reference yield (DIN, 2021), a permit for PV over farmland in Japan is in doubt when yields fall by about 20% against the local average (Ministry of Agriculture, Forestry and Fisheries of Japan, 2024), and France requires yields above 90% of a reference area (Décret n° 2024-318, 2024). Studies that design systems under such rules show that the yield limit, not the energy target, often sets the panel density (Ademollo et al., 2026). An estimate of APV potential is only meaningful, then, if it counts land where the yield condition holds and at a density that meets it.

Most potential estimates do not apply such a condition (Table 1). For India, a national study for the Indo-German Energy Forum reported 3.2 to 13.8 terawatts (TW) from threshold screening and a crop suitability matrix (Fraunhofer ISE et al., 2024), an assessment by The Energy and Resources Institute (TERI) found 1.2 to 2.1 TW on the 2.8 million hectares of cropland it judged highly suitable for shade-compatible crop categories (Yadav et al., 2026), and a geospatial multi-criteria study mapped more than 200,000 km² of highly suitable land (Mehta & Betz, 2026). None of these studies simulates crop yield under the panels. Outside India, suitability has been mapped for Great Britain without modelling crop yield (Neesham-McTiernan et al., 2025), and a global agro-hydrological study applied uniform light reductions to rainfed crops and found that 22–35% of rainfed harvested area could keep its yield (Curioni et al., 2025). The tenfold spread of the Indian estimates thus comes from assumptions about which crops tolerate shade and what power density is acceptable, not from a model of how a given array affects a given crop.

Table 1. Studies that estimate agrivoltaic potential at regional to global scale.

Study	Region and resolution	Crop yield under PV	PV geometry	Irrigation and nitrogen	Climate	Yield rule
Fraunhofer ISE et al. (2024)	India, district	Crop suitability matrix	Three configurations	No	Current	None
Yadav et al. (2026)	India, 250 m	Suitability matrix; 2.8 of 47 million hectares used	Fixed 0.42–0.75 megawatts (MW) per hectare	No	Current	None
Mehta and Betz (2026)	India, 1 km	Not modelled (multi-criteria index)	Not modelled	No	Current	None
Xia et al. (2024)	China, PV sites	Shading-response curve per crop	Two densities	No	Current	None
Willockx et al. (2022)	Europe, 25 km	Assumed unchanged; GCR from crop light need	Elevated, fixed	No	Current	Light requirement
Curioni et al. (2025)	Global, ~10 km	Agro-hydrological model, rainfed crops	Uniform light reduction	Rainfed only	Historical	None
Jia et al. (2026)	US Midwest, county	Process-based model, maize and soybean	GCR 0.33	Not specified	Current and projected, high-emission	None
Arendarczyk et al. (2026)	Global, gridded	LandSyMM (LPJ-GUESS with PLUM)	Fixed GCR 0.25	Nitrogen and irrigation simulated	Land-use scenarios to 2060	None

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Study	Region and resolution	Crop yield under PV	PV geometry	Irrigation and nitrogen	Climate	Yield rule
This study	India, 0.25°, 2977 cells	Process-based, 4 crops, calibrated to state yields	3 layouts × GCR 0.05–0.7, strip-resolved light	Irrigated and rainfed; calibrated and recommended nitrogen	1995–2014; 2041–2060 under two emission pathways, 5 climate models	Relative yield ≥ 0.66, 0.80, 0.90

GCR is the ground coverage ratio, the share of land covered by modules.

Coupled light and crop models already exist at field scale. Ground irradiance below PV rows has been computed with ray tracing (Dupraz et al., 2011) and view factors (Nassar et al., 2022), and fed to crop and water-balance models to predict yield and irrigation demand under specific arrays (Amaducci et al., 2018; Elamri et al., 2018), to optimise layout at single sites (Mazzeo et al., 2025; Ademollo et al., 2026) or to rank designs under a minimum crop ratio (Bellone et al., 2024). The approach has not been scaled to a national map, because it needs weather, soil, crop calendar and management data for every location and many crop-model runs per layout. Climate change is a further reason to use process-based models. Shade removes light, which costs yield where light limits growth, but it also lowers canopy temperature and water demand, which helps where heat and drought limit growth. In the United States Midwest the same array reduced maize and soybean yields in the humid east but raised soybean yields in the semi-arid west (Jia et al., 2026). Warming and changing rainfall may therefore move the land where APV meets a yield rule, which correlative suitability maps cannot capture. To our knowledge, no national APV potential has been tested with crop models, explicit array layouts and a yield threshold under future climate.

India is a strong test case. Its cropland spans arid, semi-arid and humid climates, and four major field crops with contrasting shade responses cover about 51 million hectares (Mha) of harvested area: wheat (28 Mha), pearl millet (9 Mha), chickpea (8 Mha) and groundnut (5 Mha) (Yu et al., 2020). Policy is also moving PV onto farms: the Pradhan Mantri Kisan Urja Suraksha evam Utthaan Mahabhiyaan (PM-KUSUM)

scheme supports ground-mounted or stilt-mounted solar plants on farmers' own land (Ministry of New and Renewable Energy, 2024). Whether such plants can keep crop yields within accepted limits, and at what density, is open.

We therefore ask how much solar capacity India's cropland can host if crops must keep a set share of their yield. A strip-resolved light model for three array layouts is coupled with Decision Support System for Agrotechnology Transfer (DSSAT) crop models for the four crops on a 0.25° grid and driven by five climate models. We ask how dense APV arrays can be while the cropland keeps 66%, 80% and 90% of its yield, how crop, layout and nitrogen management change that density and the resulting capacity, and how both shift by 2041–2060 under two Shared Socioeconomic Pathways (SSPs), SSP2-4.5 and SSP5-8.5. We also screen what keeping the crop costs, in tariff terms, against the value of the crop it protects.

2 Materials and Methods

The potential was estimated by coupling a two-dimensional light model of the ground below the PV rows with process-based crop models, for every 0.25° cell where the four crops are grown, under the current climate and two mid-century scenarios. Fig. 1 shows the workflow: the light model (A), the crop model (B), the threshold test (C) and the conversion to capacity and energy (D). Further detail is given in the Supporting Information (Text S1–S14).



Fig. 1. Methodology flowchart. (A) A light model gives the daily light factor of five ground strips below each layout and ground coverage ratio. (B) A crop model simulates open-field and shaded yields, with nitrogen calibrated by state against observed yields. (C) Relative yield is tested against each threshold τ to find the largest feasible ground coverage ratio. (D) That ratio and the eligible land give photovoltaic capacity and energy, reported as the median and range of five global climate models. Numbers in brackets refer to sections of the text.

2.1 Study domain and crop systems

Four crops cover both cropping seasons and the main contrasts in shade response: wheat (rabi, a C_3 cereal), chickpea (rabi, C_3 legume), pearl millet (kharif, C_4 cereal) and groundnut (mainly kharif, C_3 legume). Harvested area and irrigated share came from the Spatial Production Allocation Model (MapSPAM) for 2010 (Yu et al., 2020). A crop was simulated in a cell where its harvested area was at least 1000 ha, which gives

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2977 cells and keeps 93–99% of each crop's national area (Table 2). Irrigated and rainfed systems were simulated separately where each covered at least 10% of the crop's area in the cell.

Table 2. Crop systems and main crop-model settings.

Crop	DSSAT model, cultivar	Cells	Area simulated, Mha (share of national)	Irrigated share	Sowing date
Wheat	CERES-Wheat, Yecora Rojo	2095	28.2 (99%)	0.79	Spring wheat calendar
Chickpea	CROPGRO-Chickpea, JG 74	1462	8.0 (96%)	0.12	Wheat date − 14 d
Pearl millet	CERES-Millet, ICMH 451	1024	8.8 (97%)	0.05	Millet calendar; rainfed automatic
Groundnut	CROPGRO-Peanut, TMV 2	897	5.0 (93%)	0.19	Groundnut calendar; rainfed kharif automatic

DSSAT is the Decision Support System for Agrotechnology Transfer. Yecora Rojo is a short-season spring wheat; the default files contain no Indian wheat cultivar, so the state calibration (Section 2.4) also absorbs cultivar differences. Automatic sowing is triggered by soil moisture within ±15 d of the calendar date (Text S3).

2.2 Agrivoltaic layouts and light model

Three layouts were simulated (Fig. 2): an elevated south-facing array (tilt 25°), an elevated east-west ridge (tilt 10°), both with a 3 m clearance, and vertical bifacial rows facing east and west. The ground coverage ratio (GCR), module area per ground area, was varied from 0.1 to 0.5, with 0.6 and 0.7 added for wheat and pearl millet and 0.05 for the legumes, whose relative yield (RY) falls fastest. Modules were treated as opaque, and tilt and height were kept the same across India so that differences between cells reflect climate, crop and soil, not design (Table S1).

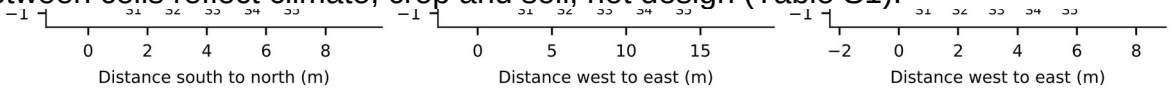


Fig. 2. Cross-sections of the three layouts at a ground coverage ratio of 0.3 (two rows shown). Blue lines are modules and grey lines supports. The cultivated row pitch is divided into five strips (S1–S5); vertical rows leave a 0.5 m uncultivated strip.

Daily radiation from the climate models was disaggregated to hourly values by scaling the Ineichen-Perez clear-sky irradiance (Ineichen & Perez, 2002) to the daily total and

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splitting it into beam and diffuse parts with the Erbs model (Erbs et al., 1982), using pvlib (Anderson et al., 2023). For each ground strip, beam transmission between the rows and the sky view factor were computed with a two-dimensional infinite-row view-factor model (Dupraz et al., 2011; Nassar et al., 2022) (Text S1). The daily light factor of strip *k* on day *d* is

(1)

where $T_k(\theta_h)$ is the beam transmission of the strip at the projected solar zenith angle θ_h of hour *h*, B_h the beam irradiance on a horizontal plane, F_k the sky view factor of the strip, D_h the diffuse horizontal irradiance and G_h the global horizontal irradiance (W m^{-2}). The crop model of each strip was driven by the open-field daily radiation multiplied by $f_{k,d}$; all other weather variables were unchanged. Against the analytical functions in pvlib, the model reproduced the mean sky view factor within 0.0005 and the hourly sunlit ground fraction within 0.0015 (Table S2).

2.3 Climate data

Daily climate came from the bias-corrected NASA Earth Exchange Global Daily Downscaled Projections (NEX-GDDP-CMIP6) at 0.25° (Thrasher et al., 2022), for five global climate models (GCMs): GFDL-ESM4, IPSL-CM6A-LR, MPI-ESM1-2-HR, MRI-ESM2-0 and UKESM1-0-LL. These are the models of the Inter-Sectoral Impact Model Intercomparison Project phase 3b (ISIMIP3b) and span equilibrium climate sensitivities of about 2.6 to 5.3 K (Meehl et al., 2020). Three 20-year periods were simulated: 1995–2014 (reference) and 2041–2060 under SSP2-4.5, an intermediate pathway, and SSP5-8.5, a high-emission pathway (Meinshausen et al., 2020). The low-emission SSP1-2.6 was not simulated because of the run time. Atmospheric CO_2 was held at the period mean of 378, 507 and 563 ppm (Meinshausen et al., 2020). Variable conversion is described in Text S2.

2.4 Crop model and calibration

Crops were simulated with DSSAT 4.8 (Jones et al., 2003; Hoogenboom et al., 2021), with Penman-Monteith evapotranspiration (Allen et al., 1998), nitrogen fixation for the legumes and the CO_2 response of each model. Sowing dates came from the Global Gridded Crop Model Intercomparison (GGCMI) Phase 3 crop calendar (Jägermeyr et al., 2021), and soil profiles to 2 m from SoilGrids 2.0 (Poggio et al., 2021) with the Saxton and Rawls pedotransfer functions (Saxton & Rawls, 2006). Irrigated systems

were irrigated automatically when soil water in the top 30 cm fell below a set fraction of plant-available water (Text S3).

Nitrogen input, which controls much of the yield gap of Indian wheat and pearl millet, differs strongly between states, and a single national rule reproduced the mean wheat yield but not its spatial pattern (Table 3). Nitrogen was therefore calibrated by state for these two crops, as is common in large-scale crop modelling (Folberth et al., 2012; Müller et al., 2017): open-field yields were simulated at 10 (pearl millet only), 30, 60, 90, 120 and 180 kg N ha^{-1} with two GCMs, and in each state the level whose area-weighted mean yield was closest to the observed 1995–2014 state yield (Swamikannu et al., 2021) was selected (Table S6). The calibrated level is an effective management intensity that also absorbs other yield-limiting factors. Across states, its Spearman rank correlation with the fertiliser nitrogen used per hectare of cropped area is 0.33 for wheat (15 states) and 0.26 for pearl millet (9 states; Text S4). The legumes were not calibrated: chickpea was unbiased on average, while groundnut was over-estimated by a factor of 2.63. Because RY is a ratio of two yields from the same model, a bias that scales both does not change it (Text S5).

Calibrated nitrogen is the base management case, and the only one used for the mid-century runs. A recommended-practice case was also run for the reference period (120 kg N ha^{-1} for irrigated wheat; Table S4). The robust case, which we use for the headline, takes in each cell the smaller GCR_{max} of the calibrated and recommended cases, so it mixes nitrogen levels by state and does not simply raise nitrogen everywhere.

Table 3. Simulated against observed open-field state yields, 1995–2014 (states with at least 50 kha of the crop).

Crop	States	Case	Pearson r	Mean absolute percentage error (%)	Simulated / observed
Wheat	15	Uncalibrated	0.01	29	0.92 (0.83)
Wheat	15	Calibrated	0.74	12	1.01 (1.01)
Pearl millet	9	Uncalibrated	0.04	31	1.01 (0.94)
Pearl millet	9	Calibrated	0.28	24	1.09 (1.05)

Crop	States	Case	Pearson r	Mean absolute percentage error (%)	Simulated / observed
Chickpea	11	Not calibrated	0.21	29	1.02 (1.01)
Groundnut	10	Not calibrated	−0.19	163	2.63 (2.36)

Mean of GFDL-ESM4 and MPI-ESM1-2-HR. The mean absolute percentage error is weighted by observed harvested area. Simulated / observed is the area-weighted mean of the state ratios; in brackets, the ratio of the area-weighted mean yields.

2.5 Yield-constrained potential

For each cell, crop system, layout, GCR and GCM, the relative yield was computed on a land-area basis, as in land equivalent ratio studies and the reference-yield rule of DIN SPEC 91434 (Dupraz et al., 2011; Weselek et al., 2019; DIN, 2021):

(2)

where $\bar{Y}_{APV}$ is the yield under the array averaged over strips and seasons, $\bar{Y}_0$ the open-field yield over the same seasons (kg ha^{-1}), and c the cultivated share of the pitch (1 for the elevated layouts). Irrigated and rainfed systems were combined by production. The largest GCR that meets a threshold τ, GCR_{max}, was found by linear interpolation at the first crossing of the RY-GCR curve, which starts at RY = 1 for GCR = 0 (Willockx et al., 2022; Dupraz, 2024); where RY stayed above τ at the largest simulated GCR, that value was used and reported as a lower bound. Three thresholds from existing rules were used: τ = 0.66 (DIN SPEC 91434 (DIN, 2021)), 0.80 (Japan (Ministry of Agriculture, Forestry and Fisheries of Japan, 2024); main case) and 0.90 (France (Décret n° 2024-318, 2024)). DIN SPEC 91434 also limits the land lost to the structure to 15%, which the 0.5 m strip of the vertical layout exceeds above GCR 0.6; vertical GCR_{max} is therefore capped at 0.6.

Eligible land was the MapSPAM harvested area of each crop multiplied by the share of the cell with terrain slope of at most 5%, the limit used to screen land for utility PV in India (Deshmukh et al., 2017), from the Copernicus 90 m elevation model (European Space Agency, 2022). Because a PV array stands all year, the rabi area A_r (wheat, chickpea) and kharif area A_k (pearl millet, groundnut) of a cell were assumed to overlap as far as possible, and the overlap had to satisfy both seasons. Every eligible hectare

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enters at its own GCR_{max}, so thresholds, layouts, management and climate change the potential through density, not land area. For a cell, capacity is

(3)

where η = 0.22 is the module efficiency, I_{STC} = 1000 W m^{-2}, $A_{ov} = \min(A_r, A_k)$ the double-cropped area, and $\bar{g}_r$ and $\bar{g}_k$ the area-weighted GCR_{max} of the rabi and kharif crops. Annual energy was obtained by multiplying capacity by the specific yield of each cell, layout and GCR, computed with pvlib for a bifacial glass-glass module using the infinite-sheds model, the Faiman cell temperature model (Faiman, 2008) and PVWatts (Dobos, 2014) (Table S8).

2.6 Uncertainty and assumptions

All steps were run separately for each GCM; national and state totals are the five-GCM median with the minimum and maximum as the range. This range is not a total uncertainty: all GCMs are bias-corrected to the same observations, and crop-model, calibration, soil and light-model errors are not propagated. Sensitivity cases change one assumption at a time (Table 4). For PV energy, daily radiation was rescaled cell by cell and month by month to the 2001–2020 NASA POWER climatology (NASA, 2026), because aerosols dim surface radiation over India (Padma Kumari et al., 2007) and reduce PV output (Bergin et al., 2017), as in linear-scaling bias correction (Teutschbein & Seibert, 2012). Three assumptions shape the results. Panels change only the radiation reaching the crop, so lower leaf temperature and evaporative demand under shade are not represented. Mid-century yields include CO_2 fertilisation, which is strongest for the C_3 legumes. And the potential is technical, bounded by crop yield and terrain only: grid capacity and land tenure are not considered, and economics enter only as a screening (Section 2.7).

2.7 Screening economics

The economic layer asks what the yield rule costs the crop and whether elevated arrays could be paid for; it is a screening estimate, not a project appraisal. For each cell, the levelised cost of electricity (LCOE) was set to the price discovered in the Gujarat reverse auction for PM-KUSUM Component A, Indian rupees (Rs) 2.77–2.78 per kWh (Gujarat Electricity Regulatory Commission, 2026), scaled by the ratio of the median specific yield of all cells to that of the cell, and a structure premium for the elevated frame of 0%, 25%, 30% or 38% was added. The Union Minister for New and

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Renewable Energy has put the extra cost of agrivoltaic plants at 25–30% (Economic Times EnergyWorld, 2026), and 38% is the excess LCOE of an elevated plant in a German analysis (Schindele et al., 2020). Electricity was valued at the regulator's ceiling tariff of Rs 2.95 per kWh (Gujarat Electricity Regulatory Commission, 2026). The gross crop value of each eligible hectare is the observed state mean yield of 1995–2014 (Swamikannu et al., 2021) times the minimum support price of each crop (Rs 2,585, 5,875, 2,900 and 7,517 per quintal for wheat, chickpea, pearl millet and groundnut (Press Information Bureau, 2025; Press Information Bureau, 2026)). The yield rule limits the crop loss to a share $(1 - \tau)$ of this value. Land lease, grid connection, financing and crop input costs were not modelled, so results are given per kWh and per hectare, not as national revenue.

3 Results

3.1 Model evaluation

Calibrating nitrogen by state improved the simulated open-field yields of the two cereals (Fig. S1). For wheat, the correlation between simulated and observed state yields rose from r = 0.01 to 0.74 and the area-weighted mean absolute percentage error fell from 29% to 12% (15 states). Pearl millet improved less, from r = 0.04 to 0.28 and from 31% to 24%, because yields in the south stayed over-predicted even at the lowest nitrogen level. The light model reproduced the pvlib view factors closely and the light measured over the whole Jodhpur system, but over-estimated the light directly below the modules under low winter sun (Figs. S2 and S3).

3.2 Feasible ground coverage ratio by crop and layout

Cereals tolerate about twice the panel density of legumes (Fig. 3A–D). In the east-west layout, the area-weighted mean GCR_{max} at RY ≥ 0.80 was 0.52 for wheat and 0.53 for pearl millet, but 0.27 for chickpea and 0.32 for groundnut. The south-facing array allowed the lowest densities for every crop, most clearly for the rabi crops wheat (0.34) and chickpea (0.19), whose grain fills under low winter sun (Fig. 3E). Vertical rows allowed densities close to the east-west layout and reached the cap of 0.6 on at least 73% of the pearl millet area. Within each crop, GCR_{max} varied strongly across the country. Maps for all crops and layouts, and their change by mid-century, are in Figs. S7–S9.

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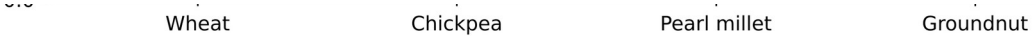


Fig. 3. Largest ground coverage ratio that keeps relative yield at or above 0.80, 1995–2014, robust nitrogen case (smaller of the calibrated and recommended nitrogen results in each cell). (A–D) East-west layout by crop: median of five climate models, interpolated for display; the number in each map is the area-weighted mean. (E) Mean by crop and layout, with error bars for the range of the five models. Boundaries: DataMeet India community, States/Admin2 (https://github.com/datameet/maps, Creative Commons Attribution 4.0 licence).

3.3 National potential

Under the yield rule, Indian cropland could host 39.2 terawatts peak (TWp) of APV (Fig. 4, Table 4). This is the east-west layout at τ = 0.80 on 39.7 Mha of cropland with slopes up to 5% (GCM range 39.0–39.8 TWp), or 0.99 megawatts peak (MWp) per hectare. It would generate 60.9 petawatt-hours (PWh) per year, about 36 times India's 2025–26 electricity requirement of 1,707 terawatt-hours (Central Electricity Authority, 2026). Seen the other way, 500 gigawatts peak at this mean density would occupy only 0.51 Mha, 1.3% of the eligible cropland. Capacity density is highest across the Indo-Gangetic plain and north-western Rajasthan and lowest in the legume-growing Deccan and the south, which follows the crop pattern of Fig. 3.

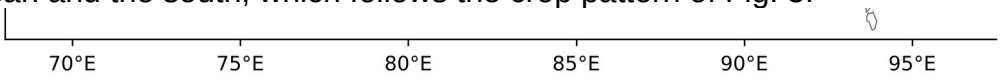


Fig. 4. Yield-constrained capacity density (megawatts peak per hectare of cropland, crops combined) for east-west elevated arrays at relative yield ≥ 0.80, 1995–2014, robust nitrogen case (smaller of the calibrated and recommended nitrogen results in each cell), slope ≤ 5%. Median of five climate models, interpolated for display; cells with less than 1000 ha of eligible cropland are blank. Labels give the capacity of the three leading states.

The layouts rank differently by capacity (Fig. 5A) and by energy (Fig. 5B). Vertical rows reached the most capacity, 41.2 TWp with the 0.6 cap, against 28.4 TWp for the south-facing array, but their specific yield was only 1,078 kWh kWp^{-1} yr^{-1} against 1,554 for east-west arrays, so they produced the least energy (44.4 PWh yr^{-1}). East-west arrays gave the most energy at every threshold and are used for the headline. The threshold mattered more than the slope rule: capacity was 47 TWp at τ = 0.66 and 31 TWp at τ = 0.90, while dropping the slope limit raised it only to 43 TWp. Three states hold 61% of the potential: Uttar Pradesh (11.8 TWp), Rajasthan (6.8 TWp) and Madhya Pradesh (5.4 TWp) (Fig. 6).

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Table 4. Yield-constrained APV potential of India for four crops, 1995–2014: robust nitrogen case and calibrated nitrogen alone (upper bound). Slope ≤ 5%; crops combined with double-cropped land counted once; vertical GCR_{max} capped at 0.6.

Layout	τ	Capacity, TWp (robust)	Capacity, TWp (calibrated)	Energy, PWh yr^{-1} (robust)	Energy, PWh yr^{-1} (calibrated)	MWp ha^{-1} (robust)
South-facing, elevated	0.66	35.7 (35.4–35.9)	40.3 (40.1–40.5)	58.5	65.0	0.90
	0.80	28.4 (27.9–28.7)	34.2 (33.9–34.3)	47.1	55.7	0.72
	0.90	21.1 (20.3–21.3)	28.7 (28.2–28.9)	35.5	47.0	0.53
East-west, elevated	0.66	47.3 (47.1–47.9)	51.1 (50.8–51.4)	72.7	77.9	1.19
	0.80	39.2 (39.0–39.8)	45.5 (45.5–45.9)	60.9	69.6	0.99
	0.90	30.9 (30.4–31.5)	39.8 (39.6–40.3)	48.5	61.0	0.78
Vertical bifacial	0.66	47.5 (47.4–47.6)	47.7 (47.6–47.8)	49.9	50.0	1.20
	0.80	41.2 (40.8–41.4)	42.8 (42.6–43.0)	44.4	45.7	1.04
	0.90	30.4 (29.4–30.8)	38.1 (37.8–38.3)	34.6	41.0	0.77

Median over five GCMs, GCM range in brackets. Land is the eligible crop area, 39.7 Mha in every case; the threshold and nitrogen act through the density. Vertical rows without the 0.6 cap (robust): 53, 43 and 31 TWp at τ = 0.66, 0.80 and 0.90.



Fig. 5. National agrivoltaic capacity (A) and annual energy (B) by layout and relative-yield threshold, 1995–2014, slope ≤ 5%. Filled bars: robust nitrogen case; outline bars: calibrated nitrogen alone (upper bound). Median of five climate models, with error bars for their range.

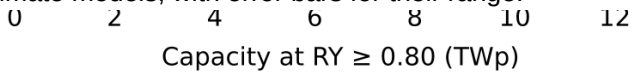


Fig. 6. Capacity of the leading states at relative yield ≥ 0.80 by layout, 1995–2014, robust nitrogen case (smaller of the calibrated and recommended nitrogen results in each cell), slope ≤ 5%. Median of five climate models, with error bars for their range.

3.4 Nitrogen management and the field-response scenario

With calibrated nitrogen alone, the east-west potential at τ = 0.80 is 45.5 TWp (GCM range 45.5–45.9) and 69.6 PWh yr^{-1}, so the robust case is 14% lower (Table 4, Fig. 5). The difference comes almost entirely from the cereals: mean GCR_{max} falls from 0.62 to 0.52 for wheat and from 0.58 to 0.53 for pearl millet, while the legumes barely change. With calibrated nitrogen, 42% of the east-west wheat area still met the threshold at

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GCR 0.7; in the robust case 1% does. These wheat cells lie mostly in states with low calibrated nitrogen, which points to nitrogen masking rather than true shade tolerance (Section 4.2). Figs. S4 and S5 show the calibrated-nitrogen maps and state results.

The crop models are also more tolerant of shade than field observations (Text S14). In the field-response scenario, simulated RY is rescaled so that its dependence on the reduction in ground light follows the observations compiled by Jha et al. (Jha et al., 2026), with maize standing in for pearl millet. The east-west potential at $\tau = 0.80$ then falls by 42% to 22.6 TWp and 36.1 PWh yr^{-1}, with GCR_{max} of 0.31 for wheat, 0.23 for pearl millet, 0.20 for groundnut and 0.17 for chickpea, and the ranking of layouts by energy is unchanged. We therefore treat the robust case as the upper bound and the field-response scenario as the lower bound of the technical potential.

3.5 Mid-century change

The climate runs use calibrated nitrogen, so the changes below are relative to the calibrated 1995–2014 potential of 45.5 TWp. By 2041–2060 under SSP2-4.5, the east-west potential at $\tau = 0.80$ changes by −1% to 45 TWp (GCM range −7% to +1%). Under SSP5-8.5 it changes by −1% to 44.9 TWp (range −6% to 0%, the paired change of each GCM against its own 1995–2014 total; Fig. 5A). At least four of five GCMs agree on the sign of the change in GCR_{max} on 56% of the eligible land under SSP2-4.5 and 57% under SSP5-8.5. Mean GCR_{max} in the east-west layout changes by +0.00 for wheat, −0.05 for pearl millet, +0.00 for chickpea and −0.01 for groundnut (Fig. 7B), and pearl millet in the north-west loses most (Fig. 7C,D).

Climate barely changes capacity but lowers energy: annual energy changes by −5% to 66.3 PWh, and specific yield by −3%. Most of this loss comes from lower irradiance in the climate models (−3.2%); the remainder (−0.3%) reflects lower module efficiency at higher temperature. These changes include the effect of higher CO_2 on the crops (Section 4.3). Change maps by crop and layout are in Figs. S8 and S9.

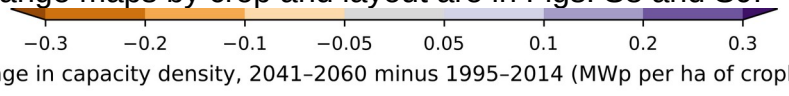


Fig. 7. Change by 2041–2060 against 1995–2014 at relative yield ≥ 0.80 with calibrated nitrogen. (A) National capacity by layout and (B) mean largest ground coverage ratio by crop, east-west layout; open symbols, SSP2-4.5 (intermediate emissions), filled symbols, SSP5-8.5 (high emissions), with the range of the five paired changes of the climate models. (C, D) Change in capacity density, east-west layout, median of five climate models, interpolated for display, for SSP2-4.5 (C) and SSP5-8.5 (D).

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3.6 Screening economics

At the ceiling tariff of Rs 2.95 per kWh, an eligible hectare under the robust east-west case would earn about Rs 45 lakh a year from electricity (1 lakh = 100,000), against a gross crop value of Rs 0.72 lakh on the same land, so electricity would be worth about 63 times the crop. The yield rule is what prevents full conversion: at τ = 0.80 the crop loss is at most Rs 569 billion a year, one fifth of the Rs 2,843 billion of gross crop value on the eligible land.

Whether the electricity can be sold at that tariff depends on the premium of the elevated frame (Fig. 8). With no premium, the energy-weighted LCOE is Rs 2.75 per kWh and 85% of the capacity (33.5 TWp) costs less than the ceiling tariff. At the ministry's 25% premium the LCOE is Rs 3.44 per kWh and no capacity clears the ceiling; at 38% it is Rs 3.80. The auction price leaves only 6% headroom below the ceiling, so a larger premium needs a tariff adder: Rs 0.49 per kWh at 25%, Rs 0.63 at 30% and Rs 0.85 at 38%.

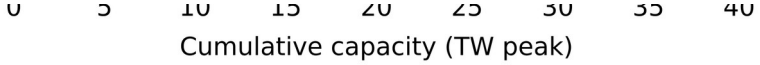


Fig. 8. Cost-supply curve of the yield-constrained agrivoltaic potential, east-west layout, relative yield ≥ 0.80, 1995–2014, robust nitrogen case, slope ≤ 5%. Cells are ordered by levelised cost of electricity. Lines are medians of five climate models and bands their range; colours show the cost premium of the elevated structure over ground-mounted photovoltaics. Horizontal lines mark the ceiling tariff and the discovered auction price of the Gujarat farmer solar scheme.

Per hectare, even the adder at a 25% premium is about Rs 8 lakh a year, many times the Rs 0.72 lakh of crop value that the yield rule protects. Crop revenue alone therefore cannot pay for elevated frames; the case for them rests on keeping land in food production (Section 4.4).

3.7 Sensitivity

Apart from the array layout, the yield threshold is the strongest lever on the potential (Table 5): moving τ from 0.80 to 0.66 or 0.90 changes capacity by +21% and −21%. Calibrated nitrogen alone changes it by +16% and dropping the slope limit by +10%, while module efficiency of 20% or 24% instead of 22% changes capacity and energy by about ±9%. Applying the threshold season by season, in at least 16 of 20 seasons, changes capacity by −14%. The spread of the five GCMs, 39.0–39.8 TWp, is small

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against all of these, so the choice of rule and management matters far more than the climate model.

Table 5. Sensitivity of the national APV potential, 1995–2014: one assumption changed at a time from the base case, with the basis of each tested value.

Case	Basis of the tested value	Capacity, TWp	Change, %	Energy, PWh yr^{-1}
Base: robust nitrogen, east-west, τ = 0.80, slope ≤ 5%	Japanese threshold (Ministry of Agriculture, Forestry and Fisheries of Japan, 2024); 5% PV slope limit for India (Deshmukh et al., 2017)	39.2	–	60.9
Threshold τ = 0.66	DIN SPEC 91434 minimum (DIN, 2021)	47.3	+21	72.7
Threshold τ = 0.90	French decree (Décret n° 2024-318, 2024)	30.9	−21	48.5
No slope limit	No terrain screening	43.0	+10	67.1
Slope ≤ 10%	Relaxed terrain limit	40.9	+4	63.7
Calibrated nitrogen alone (upper bound)	State-calibrated nitrogen, Section 2.4	45.5	+16	69.6
Module efficiency 20%	Older modules; 2025 mass production 21.7% (PERC) (VDMA Photovoltaic Equipment, 2026)	35.6	−9	55.4
Module efficiency 24%	Back-contact modules, 24.1% in 2025 (VDMA Photovoltaic Equipment, 2026)	42.8	+9	66.5
South-facing layout	Alternative design, Section 2.2	28.4	−28	47.1
Vertical layout, GCR ≤ 0.6	Alternative design, Section 2.2	41.2	+5	44.4
Vertical layout, no DIN cap	15% land-loss limit removed (DIN, 2021)	43.0	+10	45.7
Threshold met in ≥ 16 of 20 seasons	Rules judge single seasons (DIN, 2021; Décret n° 2024-318, 2024); one failure in five tolerated	33.8	−14	52.9
Radiation not corrected (energy only)	NEX-GDDP radiation as supplied, Supplementary Section S12	39.2	0	70.4

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Case	Basis of the tested value	Capacity, TWp	Change, %	Energy, PWh yr^{-1}
Range of five GCMs (base case)	CMIP6 ensemble, Section 2.3	39.0–39.8	–	–

Five-GCM median. Module efficiency scales capacity and energy in proportion, because GCR_{max} depends on shade, not on the module rating.

4 Discussion

4.1 The yield-constrained potential against earlier estimates

In the robust case (east-west layout, τ = 0.80, slope ≤ 5%), the four crops can host 39.2 TWp of APV (GCM range 39.0–39.8) on 39.7 Mha of eligible cropland. This is 90% of the land the four crops occupy before the slope rule and 79% of their summed harvested area (Fig. 9). Calibrated nitrogen alone gives 45.5 TWp, which we treat as an upper bound because low nitrogen input masks part of the shade loss in wheat (Section 4.2), and rescaling the shade response to field observations lowers the robust figure to 22.6 TWp (Section 3.4). The technical potential therefore lies between 22.6 and 39.2 TWp.

The figure is 19–33 times the 1.2–2.1 TW of TERI (Yadav et al., 2026) and 2.8 times the upper end of the 3.2–13.8 TW reported for the Indo-German Energy Forum (Fraunhofer ISE et al., 2024), but the studies do not measure the same thing. TERI converts only the 2.8 million hectares of its 47 million hectares of restricted cropland that it judged highly suitable for shade-compatible crop categories, and the Fraunhofer study assigns suitability from a crop matrix and screening thresholds. Our larger figure shows what the yield rule alone permits on the land of four crops, before grid, economic or social limits, and is not evidence that the earlier studies were too cautious. Two differences explain most of the gap. The first is the land base: TERI's fixed density of 0.42–0.75 MW ha^{-1} applied to our 39.7 Mha would already give 17–30 TW. The second is the power density. With η = 0.22, TERI's range equals a GCR of 0.19–0.34, whereas our yield-constrained density of 0.99 MWp ha^{-1} is 1.3–2.4 times higher and not uniform: GCR_{max} ranges from 0.20 to 0.60 across the eligible area (5th to 95th percentile). A fixed density is thus too high for legume land and too low for much of the irrigated wheat belt.

Fig. 9 follows the gap on the same grid and crops. A fixed 0.75 MW ha^{-1} on all simulated crop area gives 37.5 TW (0.42 MW ha^{-1}: 21.0 TW). The slope rule and

counting double-cropped land once remove 21% of the area and leave 29.8 TW at the higher density. Replacing the fixed density by GCR_{max} raises capacity by +53% against 0.75 MW ha^{-1} and by +173% against 0.42 MW ha^{-1}. Robust nitrogen management then changes the result by −14%. The field-response lower bound (22.6 TWp) lies between the two fixed-density cases on the same land, so the choice of shade response matters about as much as the choice of fixed density.

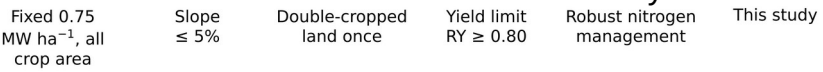


Fig. 9. From a fixed power density to the yield-constrained potential, east-west layout, relative yield ≥ 0.80, 1995–2014. Steps are applied cumulatively on the same grid and crops. Bars use 0.75 megawatts per hectare for steps 1–3 and diamonds 0.42 (the range of an Indian national assessment (Yadav et al., 2026)). Error bars are the range over five climate models. The last step uses the robust case (minimum of calibrated and recommended nitrogen).

Elsewhere, spatial studies for Great Britain, Europe, Phoenix, the Middle East and North Africa and the globe count suitable land under assumed layouts or shading levels (Majumdar & Pasqualetti, 2018; Willockx et al., 2022; Curioni et al., 2025; Neesham-McTiernan et al., 2025; Abdalla et al., 2026). The global study applied uniform light cuts of 10–50% to rainfed crops and found that 22–35% of rainfed harvested area could be converted with limited yield loss, naming parts of India as hotspots (Curioni et al., 2025). We add array layouts, irrigated crops, nitrogen, a yield threshold and future climate on a national grid. For India, a crop-model study of two array densities reported land equivalent ratios of 1.35–1.97 (Aswal et al., 2026), which show a gain in land productivity but set no minimum crop yield. The closest study in design is a Japanese assessment that optimised layouts under daily-light constraints and found a mean GCR of 30% (Nakata & Ogata, 2026), within our range (0.20–0.60).

4.2 What sets the panel density: nitrogen, crop and layout

Management mattered as much as the crop. In the Rajasthan pilot (Text S9; 498 cells, five GCMs, 1995–2014), wheat under the south-facing array at GCR 0.3 had a median RY of 1.13 at the farmer rate of 60 kg N ha^{-1} and 0.90 at the recommended 120 kg N ha^{-1}. RY above one is not a real yield gain. The likely mechanism is nitrogen dilution: the critical nitrogen concentration of a crop falls as its biomass grows (Justes et al., 1994), so a shaded crop that grows less stays closer to its nitrogen demand. Under low

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nitrogen the open-field crop is limited by nitrogen and the shaded crop by light, and the light loss is hidden.

The national calibrated results show the same signature. 42% of the east-west wheat area kept RY ≥ 0.80 even at GCR 0.7, and this share follows the calibrated nitrogen level: 67% in states calibrated at 30 kg N ha^{-1} and 59% at 60, but 10% at 90 and 4% at 180. In the wheat cells at the limit the open-field yield is only 34% of the yield simulated at 180 kg N ha^{-1}, against 61% elsewhere, so much of this tolerance reflects nitrogen-limited open-field yields and not shade-tolerant crops. Management is also not fixed: a farmer who receives lease income could apply more fertiliser, and the rule would then become stricter at the same density. We therefore take the smaller GCR_{max} of the calibrated and recommended cases as the headline. This is a conservative scenario rule, not a probability, and it cuts the national potential by 14% (from 45 to 39 TWp), with the largest absolute losses in Uttar Pradesh and Madhya Pradesh. Irrigation, sowing date and cultivar could also shift tolerance, and we did not vary them.

Among crops, the ranking agrees with the meta-analysis of Laub et al. (Laub et al., 2022), which found strong losses in grain legumes and maize even at low shade and smaller losses in C_3 cereals, and with a later synthesis of 546 observations (Jha et al., 2026). Chickpea (0.27) and groundnut (0.32) tolerate about half the panel density of wheat (0.52) and pearl millet (0.53) in the east-west layout. The two legumes hold 25% of the eligible crop area but 15% of the capacity at $\tau = 0.80$, and where a legume and a cereal share a field in the two seasons the legume usually sets the density. Pearl millet is the result that needs most care. The global study found that C_4 crops such as millet respond less well to reduced radiation (Curioni et al., 2025), and the meta-analyses found strong losses for maize. Yet a stilt-mounted system in Japan with low module density raised maize yield per square metre by 5.6% (Sekiyama & Nagashima, 2019). Rainfed pearl millet at the calibrated 10–30 kg N ha^{-1} is limited by nitrogen and water more than by light, the same masking as in wheat, so its GCR_{max} should be read as tolerance under current low-input management and not as a property of the crop.

Layout mattered most for the rabi crops. The south-facing array, tilted at 25°, casts long shadows in winter when the sun is low, which is when wheat and chickpea fill grain: its GCR_{max} was 0.34 for wheat and 0.19 for chickpea, against 0.52 and 0.27 in the east-west layout. Vertical rows spread the shade evenly but lose cropland to the structure: the yield rule alone would exceed 0.6 on 40% of the vertical-layout area, so

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we cap it there, which lowers vertical capacity by only 4%. Vertical rows still reach the most capacity but, with their low specific yield, deliver the least energy, so a layout is better judged by its energy than its peak capacity.

4.3 Climate change and the CO_2 caveat

The yield-constrained capacity changes little by mid-century: −1% under SSP5-8.5 and −1% under SSP2-4.5 (Section 3.5). Because RY is a ratio, warming that lowers open-field and shaded yields alike leaves it unchanged, and only the part of the climate signal that differs between sun and shade moves the threshold. Light-only shading cannot represent the main such difference, the cooler and less evaporative environment under the panels (Section 4.5). Annual energy falls more, by −5% under SSP5-8.5, mostly through lower irradiance in the climate models (−3.2%), in line with projected declines in surface solar radiation over India of 0.52–0.60% per decade (Jadhav & Bhawar, 2025). Absolute yields are a different matter. Mid-century runs include CO_2 at 507 and 563 ppm, and free-air enrichment experiments show smaller yield gains for C_3 crops than the enclosure studies on which earlier model factors were based (Ainsworth & Long, 2005; Long et al., 2006), so mid-century production figures, especially for chickpea and groundnut, are likely to be optimistic. The effect on RY is much smaller, because CO_2 raises the yield under the array and in the open field alike. Under SSP5-8.5 the median open-field yield changed by +45% for chickpea and +32% for groundnut, yet the median RY at GCR 0.3 changed by only +0.01 and 0.00. The CO_2 share of these changes was not separated, because no run with fixed CO_2 was made.

4.4 Food, energy and farmers: implications for India

The modelled land resource is large relative to demand: the robust-case energy is about 36 times the national electricity requirement of 2025–26 (Central Electricity Authority, 2026) (about 21 times in the field-response scenario). This is an energy-equivalence benchmark, not a supply estimate. The main message is that the yield rule limits how densely panels can be built on farmland far more than it limits where, so the food–energy trade-off can be managed through density and crop choice. Siting near substations, grid capacity, finance, land tenure and farmer acceptance are not modelled and will decide how much of the resource can be used. Who bears the cost is also open. A farmer on land that meets $\tau = 0.80$ keeps at least 80% of the open-field

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yield on average, but whether lease income covers the remainder depends on prices, tenure and who owns the structure. In a German case study the levelised cost of APV electricity was 38% above ground-mounted PV, and the price-performance ratio was favourable for potato but not for winter wheat, so the authors advised against APV in crop rotations (Schindele et al., 2020). Our screening estimate (Section 3.6) points the same way. At the Indian ministry's premium of 25–30% the elevated frame needs a tariff adder of Rs 0.49–0.63 per kWh above the Rs 2.95 ceiling of PM-KUSUM (Rs 0.85 at the German 38%), about Rs 8 lakh per hectare a year against Rs 0.72 lakh of crop value. The adder would therefore be justified by land and food policy, not by the crop income it protects, and it would have to be tied to the yield rule so that it pays only for arrays that keep the crop.

India's main route for PV on farmland is component A of PM-KUSUM, which supports grid-connected plants of 0.5–2 MW on farmers' land, either ground-mounted or on stilts (Ministry of New and Renewable Energy, 2024; Ministry of New and Renewable Energy, 2026a). To our knowledge the scheme sets no yield rule for the crop below, whereas Germany, Japan and France do, with thresholds of 66%, about 80% and 90% of a reference yield (DIN, 2021; Décret n° 2024-318, 2024; Ministry of Agriculture, Forestry and Fisheries of Japan, 2024). The choice among them is not a detail: moving from $\tau = 0.66$ to 0.90 changes the national potential from 47 to 31 TWp. Three elements could be adopted: a relative yield threshold, with 0.80 as a defensible middle value; a limit on land lost to the structure, as in DIN SPEC 91434; and a reference that farmers and inspectors can check, for which district yields from official statistics (Swamikannu et al., 2021) could serve where no control plot exists. A crop-specific density cap, low for pulses and higher for cereals, would be simpler still and follows from the GCR_{max} maps. Component A has a target of 10 GW (Ministry of New and Renewable Energy, 2024), which at the densest cells of our map would occupy about 65 km^2, or 0.02% of the eligible land, so land area alone does not limit the scheme. The potential is also concentrated: five states (Uttar Pradesh, Rajasthan, Madhya Pradesh, Haryana and Punjab) hold 75% of the east-west capacity at $\tau = 0.80$, mostly on wheat, the most shade-tolerant of the four crops, so a yield rule would direct plants toward land where the crop tolerates the most shade.

4.5 Limitations and next steps

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The spread across the five climate models is small (39.0–39.8 TWp) and says little about the real range, which comes mainly from four choices: the shade response (the field-response scenario gives 22.6 TWp, 42% below the robust case), the yield threshold ($\tau = 0.66$ gives 47 TWp and $\tau = 0.90$ gives 31 TWp), nitrogen management (45 TWp with calibrated nitrogen alone) and the season rule (34 TWp when 16 of 20 seasons must meet the threshold). The threshold and nitrogen are policy and management choices more than scientific unknowns, which is why we report them as scenarios and not as error bars. The economic layer is a screening estimate: it scales one auction price by specific yield, takes the frame premium from a ministerial statement and a German study rather than measured Indian projects, and omits lease, grid and financing costs, so its conclusion is the order of magnitude of the adder and not a tariff to adopt.

The shade response has been checked against published observations only, and the check does not favour the model. The meta-analyses agree with the simulated ranking (Laub et al., 2022; Jha et al., 2026), but simulated RY is higher than observed. Against the 38 wheat, 97 grain-legume and 66 maize observations compiled by Jha et al. (Jha et al., 2026), the mean bias with recommended nitrogen is +0.09 for wheat, +0.09 for grain legumes and +0.20 for maize, the proxy for pearl millet (Figs. S10 and S11, Text S14). The comparison is imperfect: most observations come from temperate shade-net or intercropping trials, the compilation holds no pearl millet and none of it is from India, and warmer, higher-radiation sites keep yields better under shade, which favours more tolerance in India. Even so, the direction is consistent, and the model is probably too tolerant of shade. Rescaling it to the observations lowers the east-west potential to 22.6 TWp, with GCR_{max} of 0.31 for wheat and 0.23 for pearl millet, the latter resting on maize as a proxy. The truth probably lies between the two scenarios. Where field data on wheat shaded during grain filling exist, they warn of more than proportional losses: 20%, 50% and 80% shade cut grain yield by 27.6%, 49.0% and 60.2% in north China (Guo et al., 2010). Indian field trials of pearl millet, chickpea and groundnut under panels would settle this.

Shading was represented as a change in daily light only, a limitation shared by current crop models in APV studies (Zainali et al., 2025). Field studies report little change in mean crop temperature under panels in a temperate climate (Marrou et al., 2013) but yield gains in hot, dry seasons (Barron-Gafford et al., 2019; Weselek et al., 2021), and

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a process-based study found that shade can raise yields where aridity dominates (Jia et al., 2026). Omitting the microclimate is therefore likely to make RY conservative for rainfed crops in the arid north-west. Diffuse light and the alternation of sunflecks and shade within a day are also not represented, because the crop models receive only daily totals. The threshold was applied to RY of the 20-year mean yields, whereas rules such as DIN SPEC 91434 check individual seasons (DIN, 2021); requiring RY ≥ 0.80 in at least 16 of 20 seasons lowers the potential by 14% to 34 TWp (Table 5). The crop simulations have further limits: wheat was simulated with a short-season spring cultivar, groundnut yields were over-estimated by a factor of 2.63, and simulated yields can be as sensitive to soil data as to climate (Folberth et al., 2016). Only four crops and two emission pathways were run, and the potential is technical in the usual sense of energy studies (Lopez et al., 2012): it ignores grid capacity, land tenure, finance and farmer acceptance. Next steps are field trials in India, crop models that carry the panel microclimate, and a robust-nitrogen run for 2041–2060.

5 Conclusions

This study mapped the agrivoltaic potential of India under a crop-yield rule, by coupling a strip-resolved light model with calibrated DSSAT crop models for four field crops, three layouts and five climate models.

A yield rule mainly limits how densely solar can be built on farmland, not where, so the trade-off between food and energy can be managed through density and crop choice. On cropland with slopes up to 5%, east-west elevated arrays that keep 80% of the open-field yield could reach 39.2 TWp on 39.7 Mha, about 0.99 MWp ha^{-1}, with the shade response of the crop models, and 22.6 TWp when that response is rescaled to field observations. Both figures lie above the published Indian estimates, which exclude field crops or fix a low density. They are land-based technical potentials, not deployment forecasts: grid access, finance, land tenure and farmer acceptance, which were not modelled, will decide how much can be used. The tolerable density depends strongly on the crop: 0.52 for cereals against 0.29 for legumes, so a single national density rule would be too high for pulses and too low for much of the wheat belt.

Nitrogen management changes the answer. Where the open-field crop is held back by low nitrogen, shade costs little extra yield and the model overstates shade tolerance. Taking the lower GCR_{max} of the calibrated and recommended nitrogen runs removes

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14% of the calibrated potential and almost all wheat that seemed to tolerate the densest arrays (42% of the area before, 1% after). Mid-century climate barely changes the yield-constrained capacity (−1% under SSP5-8.5, −1% under SSP2-4.5) but changes annual energy by −5%, mostly through lower irradiance in the climate models; these results include CO_2 fertilisation, which raises absolute legume yields far more than relative yield.

For policy, India could add to PM-KUSUM a relative-yield threshold of about 0.80, a limit on land lost to the structure and a reference yield that can be checked. A screening of costs shows that elevated frames would need a tariff adder of Rs 0.49–0.85 per kWh at a structure premium of 25–38%, far above the crop value it protects, so the adder should be tied to the yield rule. Future work should test the shade response in Indian field trials, add the microclimate under the panels, test yield rules season by season, and replace the screening costs with project data and grid limits.

Author Contributions

Mohamed Al-Mandhari: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. Aritra Ghosh: Conceptualization, Methodology, Supervision, Project administration, Funding acquisition, Writing – review & editing.

Conflict of Interest Statement

The authors declare there are no conflicts of interest for this manuscript.

Acknowledgments

[Funding information for both authors, and any thanks, to be added. If there is no funding, state that no specific funding supported this work.]

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[The simulation outputs behind the figures and tables (cell-level capacity, relative yield and economics tables) will be deposited in a public repository, with its DOI cited here and in the reference list. Software availability statement to be added if the editor requires it.] Input data are public: NEX-GDDP-CMIP6 (Thrasher et al., 2022), MapSPAM (Yu et al., 2020), SoilGrids 2.0 (Poggio et al., 2021), the Copernicus digital

elevation model (European Space Agency, 2022) and the ICRISAT district database (Swamikannu et al., 2021).

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